

# Training Data for the DNO Surrogate

Wave families, depth coverage, and rescaled test sets

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## Recap

The depth-conditioned FNO was previously trained on 200k samples: Tanaka solitons at  $h = 1$  and random sea at  $h = 1000$ . Evaluation used legacy Stokes, soliton, and linear test sets.

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The depth-conditioned FNO was previously trained on 200k samples: Tanaka solitons at  $h = 1$  and random sea at  $h = 1000$ . Evaluation used legacy Stokes, soliton, and linear test sets.

This deck describes the current training data, and how the legacy test sets fit into the same coordinate system.

$$(\eta, \xi, h) \longmapsto \mathcal{G}(\eta; h)\xi$$

- Surface elevation  $\eta$
- Surface velocity potential  $\xi$ , consistent with  $\eta$  via the Laplace BVP
- *Depth*  $h$

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The dimensionless depth  $kh$  separates regimes: deep  $kh \gg 1$ , finite  $kh \sim 1$ , shallow  $kh \ll 1$ . Training data must span all three.

## Five wave families

- Random sea: broadband, phase-randomized, multi-scale
- Stokes wavetrains: analytic, narrowband, sharp-crested
- Linear wavetrains: single trig mode
- Tanaka solitons: localized, large-amplitude, shallow
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Each family is generated at multiple depths.

## The combined dataset

| source            | samples | family        | regime         |
|-------------------|---------|---------------|----------------|
| random_sea_deep   | 241k    | broadband     | deep           |
| random_sea_finite | 220k    | broadband     | finite         |
| linear            | 500k    | single mode   | shallow-deep   |
| stokes_deep       | 500k    | Stokes-5      | deep           |
| stokes_finite     | 500k    | Stokes-5      | finite-shallow |
| tanaka            | 2.0M    | solitary      | shallow        |
| bf                | 2.0M    | Benjamin-Feir | deep           |
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Single shuffled file `combined_dataset_v3.npz` on  $[0, 2\pi]$ ,  $N_x = 1024$ . Metadata:  $(\eta, \xi, \mathcal{G}\xi, h, t, \text{data source})$ .

## Random parameters per family (part 1)

| family               | parameter               | distribution                                                          |
|----------------------|-------------------------|-----------------------------------------------------------------------|
| <b>linear</b>        | $n_0$                   | $U\{1, \dots, 20\}$                                                   |
|                      | $a_0$                   | $U[10^{-4}, 1.15 \cdot 10^{-2}]$ , reject on $\varepsilon$            |
|                      | $\varepsilon = k_0 a_0$ | $[5 \cdot 10^{-3}, 0.15]$                                             |
|                      | $h$                     | $\log U[0.02, 1.5]$ , $kh \in [0.02, 20]$                             |
|                      | phase, direction        | $U[0, 2\pi]$ , $\{\pm 1\}$                                            |
| <b>stokes_deep</b>   | $n_0$                   | $U\{1, \dots, 20\}$                                                   |
|                      | $a_0$                   | $U[7.66 \cdot 10^{-4}, 1.15 \cdot 10^{-2}]$ , $\varepsilon \leq 0.15$ |
|                      | $h$                     | $\log U[4, 50]$                                                       |
|                      | $t$                     | $U[0, 20]$                                                            |
| <b>stokes_finite</b> | $n_0$                   | $U\{14, \dots, 26\}$                                                  |
|                      | $a_0$                   | $U[7.66 \cdot 10^{-4}, 1.15 \cdot 10^{-2}]$ , $\varepsilon \leq 0.15$ |
|                      | $h$                     | $\log U[0.02, 1.5]$ , $kh \in [0.5, 5]$                               |
|                      | $t$                     | $U[0, 20]$                                                            |

## Random parameters per family (part 2)

| family     | parameter                           | distribution                            |
|------------|-------------------------------------|-----------------------------------------|
| random sea | $H_s$                               | $U[0.1, 0.6]$                           |
|            | $k_p$                               | $U[0.06, 0.50]$                         |
|            | bandwidth $\gamma$                  | $U[1.5, 5.0]$                           |
|            | $h$                                 | $\log U[h_{\min}, h_{\max}]$ per regime |
|            | Fourier phases                      | i.i.d. $U[0, 2\pi]$                     |
| Tanaka     | $n_{\text{crests}}$                 | $U\{n_{\min}, \dots, n_{\max}\}$        |
|            | per-crest weights                   | Dirichlet( <b>1</b> )                   |
|            | centers, directions                 | uniform / min-sep; $\{\pm 1\}$          |
|            | $h$                                 | $\log U[0.01, h_{\max}]$                |
| BF         | $n_{\text{carr}}$                   | $U\{4, \dots, 20\}$                     |
|            | $\varepsilon_{\text{carrier}}$      | $U[0.05, 0.13]$                         |
|            | offset, $\varepsilon_{\text{pert}}$ | $U\{2, 3, 4\}, U[0.05, 0.20]$           |
|            | $h$                                 | $\log U[0.5, 4.0]$                      |

Time-evolution datasets: each draw rolls out the Euler equations in time, and saves  $K$  saved snapshots. We use adaptive sampling for the *Tanaka* and *BF* datasets. We use uniform sampling for the random sea dataset.

## Stokes-5 wavetrains

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For `stokes_finite` the  $kh \geq 0.5$  floor avoids the  $1/\sinh(kh)^7$  singularities in the Fenton coefficients. Steepness  $\varepsilon = k_0 a_0 \leq 0.15$ .

Single-mode  $(\eta, \xi)$  at the linear dispersion relation, with the MATLAB  $\xi$  including the  $\cosh(k_0(\eta + h))/\cosh(k_0 h)$  free-surface factor:

$$\eta = a_0 \cos(k_0 x + \phi), \quad \xi = \pm \frac{a_0 g}{\omega} \cdot \frac{\cosh(k_0(\eta + h))}{\cosh(k_0 h)} \sin(k_0 x + \phi).$$

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Linear theory has no asymptotic-validity restriction and covers the  $kh \ll 1$  tail that Stokes cannot reach.

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A deep-water Stokes carrier with a small sideband perturbation, integrated for hundreds of carrier periods. The trajectory exhibits the full modulational instability cycle: sideband growth, energy concentration, recurrence.

Provides deep-water large-amplitude narrowband data. Random sea is broadband, Stokes has no envelope evolution, so neither covers this regime.

## Adaptive temporal sampling (Tanaka, BF)

For time-evolving families we roll out the Euler equations on a dense grid ( $\Delta t = 0.08$ ,  $t_{\max} = 200$ ,  $T = 2501$ ) but only save  $K = 200$  snapshots per case. Uniform sampling of time snapshots do not well represent the events that are most important in each rollout: soliton collisions in Tanaka, recurrence in BF.

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**Per-case density.** For each trajectory  $b$ , define the following probability distribution by the density:

$$\text{density}_b(t) = (1 - \alpha) \text{uniform}(t) + \alpha \cdot \text{smooth}_\sigma(\text{activity}_b(t)^p),$$

and  $K$  snapshot indices are drawn via deterministic inverse-CDF at the  $K$  equispaced quantiles.

## Activity signals

Use *non-conserved* scalar values as our activity functions:

- **Tanaka (rate).**  $S(t) = \|\eta_x\|_2^2$ , activity =  $|dS/dt|/(S + \varepsilon)$ .  $|dS/dt|$  is large on both sides of a collision  $\Rightarrow$  symmetric clustering around the event.
- **Benjamin-Feir (magnitude).**  $M(t) = \|\eta\|_\infty$ , activity =  $(M - \min_t M)^p$ . Envelope modulates  $\sim 20\%$ , so  $p = 4$  locks onto recurrence humps.

|                          | Tanaka           | BF                     |
|--------------------------|------------------|------------------------|
| signal                   | $\ \eta_x\ ^2$   | $\ \eta\ _\infty$      |
| density mode             | rate $ dS/dt /S$ | magnitude $M - \min M$ |
| $\alpha$ (bias)          | 0.5              | 0.85                   |
| $\sigma$ (smooth, steps) | 50               | 20                     |
| power $p$                | 1.0              | 4.0                    |

Random sea is now generated at variable depth:

- Deep:  $h \sim \mathcal{U}_{\log}[5, 25]$
- Finite:  $h \sim \mathcal{U}_{\log}[0.5, 5]$

## Random sea and Tanaka updates

Random sea is now generated at variable depth:

- Deep:  $h \sim \mathcal{U}_{\log}[5, 25]$
- Finite:  $h \sim \mathcal{U}_{\log}[0.5, 5]$

Tanaka now uses multi-crest initial conditions, 1–3 crests, with the composite  $\mathcal{G}$  recomputed nonlinearly. Two gauge variants `tanaka_g0/g1` contribute 1 M samples each.

## Coverage in $(k_p, h)$

- Tanaka: shallow, large-amplitude, localized
- Linear very-shallow:  $kh \ll 1$
- Stokes finite: moderate  $kh$
- Random sea finite, deep: broadband mid and deep
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The training distribution is the union of these regions.

## Mapping legacy test sets to the canonical domain

The reference test set is on  $[0, 164]$ . Map it to  $[0, 2\pi]$  via the Zakharov-invariant rescaling under  $(x, y, t) \rightarrow (\alpha X, \alpha Y, \sqrt{\alpha} \tau)$  with  $\alpha = 164/(2\pi) \approx 26$ :

$$\eta_{\text{new}} = \frac{\eta_{\text{old}}}{\alpha}, \quad \xi_{\text{new}} = \frac{\xi_{\text{old}}}{\alpha^{3/2}}, \quad \mathcal{G}_{\xi_{\text{new}}} = \frac{\mathcal{G}_{\xi_{\text{old}}}}{\alpha^{1/2}}, \quad h_{\text{new}} = \frac{h_{\text{old}}}{\alpha}.$$

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Powers from  $[\eta] = L$ ,  $[\xi] = L^2/T$ ,  $[\mathcal{G}\xi] = L/T$  with  $g$  preserved. The static Laplace BVP closes under any common factor on  $(\xi, \mathcal{G}\xi)$ ; the Zakharov powers preserve numerical magnitudes between old and new domains.

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The legacy stokes set mixes deep ( $h_{\text{use}} = 1000$ ) and finite-depth ( $h = 1$ ) samples. Each subset is rescaled with its own  $h_{\text{old}}$ .

## Out-of-range depth at test time

After rescaling, deep-water test cases sit at  $h_{\text{new}} \approx 38$ . Training depths span  $[0.01, 50]$  (stokes\_deep up to 50).

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**Saturating depth clip** applied inside the model:

$$h_{\text{enc}} = \min(h, h_{\text{sat}}), \quad h_{\text{sat}} = 5.$$

The threshold matches the slide's deep-water criterion: at  $k_{\text{min}} = 1$ ,  $\tanh(1 \cdot 5) \approx 0.99991$ . All  $h \geq 5$  samples (stokes\_deep training, rescaled deep-water tests) share one depth encoding; finite-depth families ( $h \leq 4$ ) are untouched.

# Summary

- Five wave families across shallow, finite, and deep regimes
- One canonical domain  $[0, 2\pi]$ , one combined  $\sim 5$  M-sample dataset
- Legacy test sets ported via Zakharov-invariant rescaling, with per-subset  $h_{\text{old}}$
- Depth feature clipped at  $h_{\text{sat}} = 5$  inside the model so all deep-water cases (training + test) share one encoding
- Open work: getting the model converged and accurate across all sources